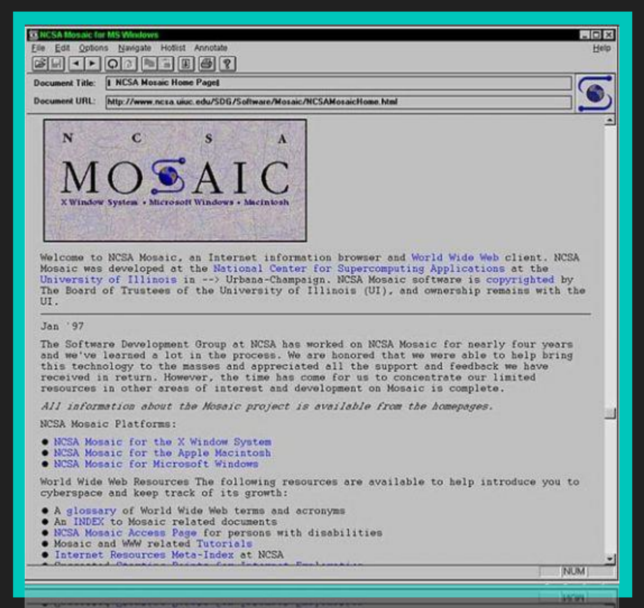
!knot: Bringing the Web Browser to the 21st Century

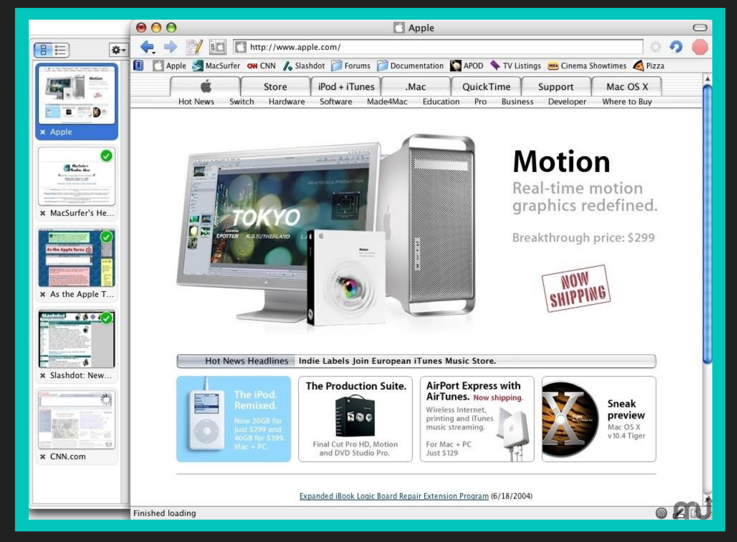
Sean Godard

Mentor: Dr. Choong-Soo Lee



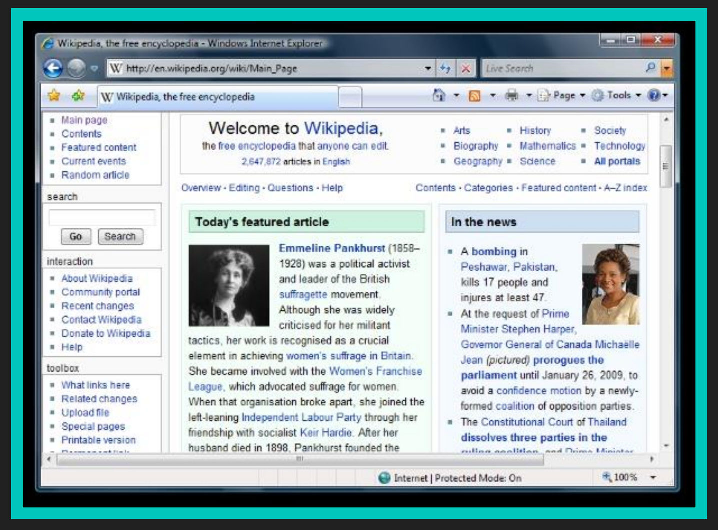
1993
Mosaic 1.0

Developed by the National Center for Supercomputing Applications, Mosaic was the first web browsers to popularize the internet to the general population and included many of the same features of today's browsers, such as the ability to bookmark websites.[8] At this time, however, users needed to open and new window for each new page they would like to view.[7]



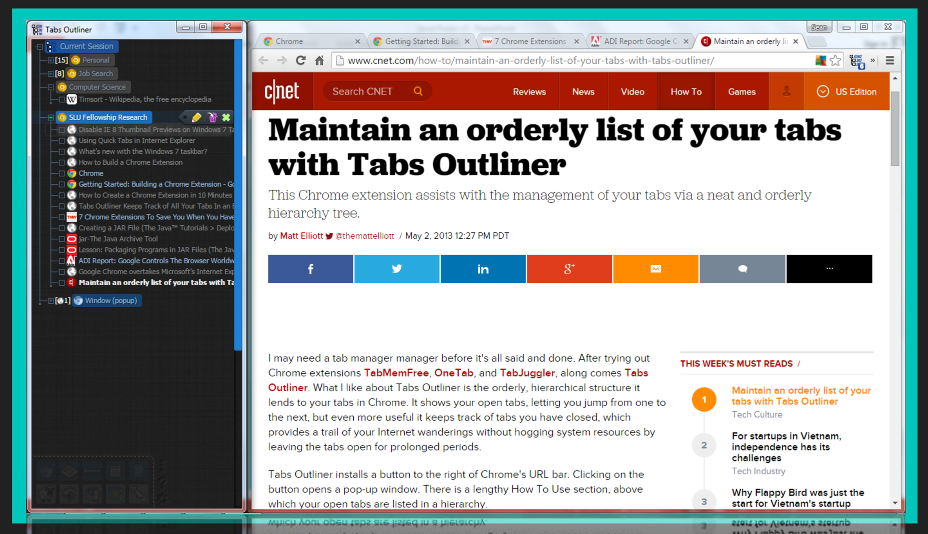
2004
Tab Preview

With the release of the OmniWeb browser, users saw the first attempt at a thumbnail image-based tab system. Compared to tabs that use text, this feature allowed for visual identification of tabs based on their content instead of labels.[9]



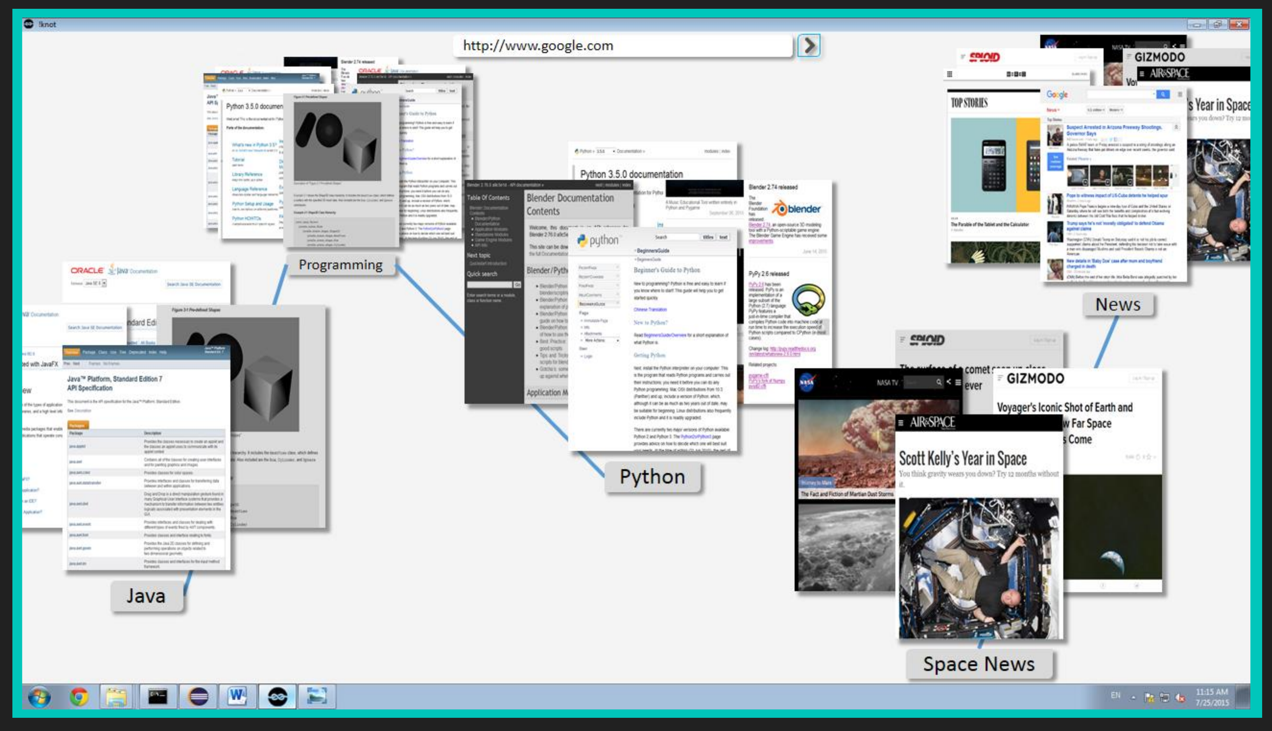
2007
Internet Explorer 7.0

While the idea of using tabs in a GUI existed long before web browsers[10], it wasn't until the mid 1990s that browsers such as NetCaptor began using this interface idea[3]. By the release of Internet Explorer 7.0, all mainstream browsers of the time included the tab system[5], allowing users to open multiple pages within the same window.



2012
'Tabs Outliner' for Chrome

A Chrome Extension in development by Vladyslav Volovyk in an attempt to solve the disorganized mess the tab system can cause. Features include the ability to organize tabs with notes in a tree-like structure and the ability to close a tab while still saving it to the tree.[11] While this is a useful tool, it is not integrated into the web browser and lacks tab previews so users still must identify tabs by their headers alone.



TBD
!knot

The final version of !knot will include all gestures of the current version in addition to features such as history, bookmarking, automatic and manual document and group linking, a save feature, selectively closing pages that can be recalled later, quick carouseling through nodes in a group, and quick access panel to dive deeper into groups and subgroups. (Concept image shown)

The Problem

Despite increasing internet usage and the fact that the tab system currently in use by most web browsers can lead to disorganization, the web browser has undergone little changes to its main user interface.

Tools

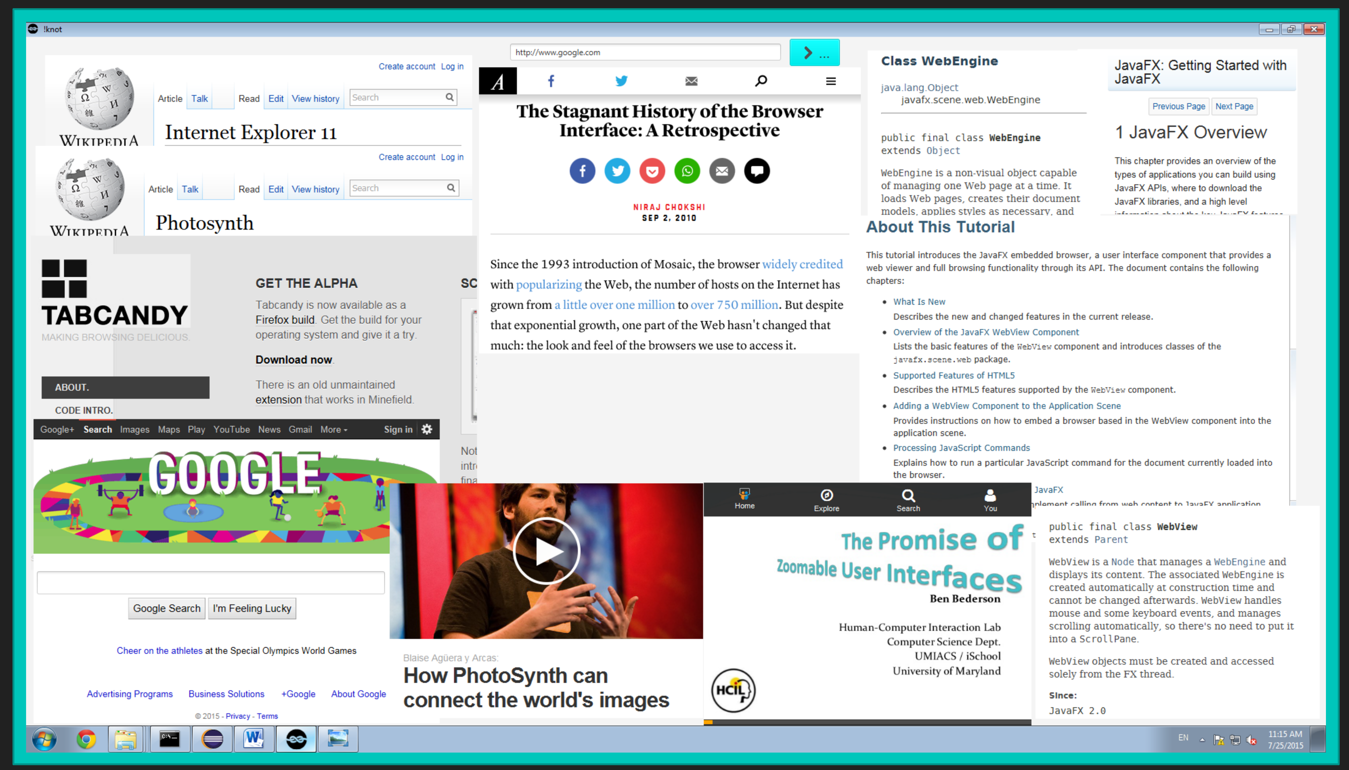
JavaFX – an application programming interface used to create graphical user interfaces for Java based applications which is also capable of rendering web content[6]

Launch4j – a Java application wrapper that bundles Java applications with the Java Runtime Environment to ease the setup process for use of java application to users[5]

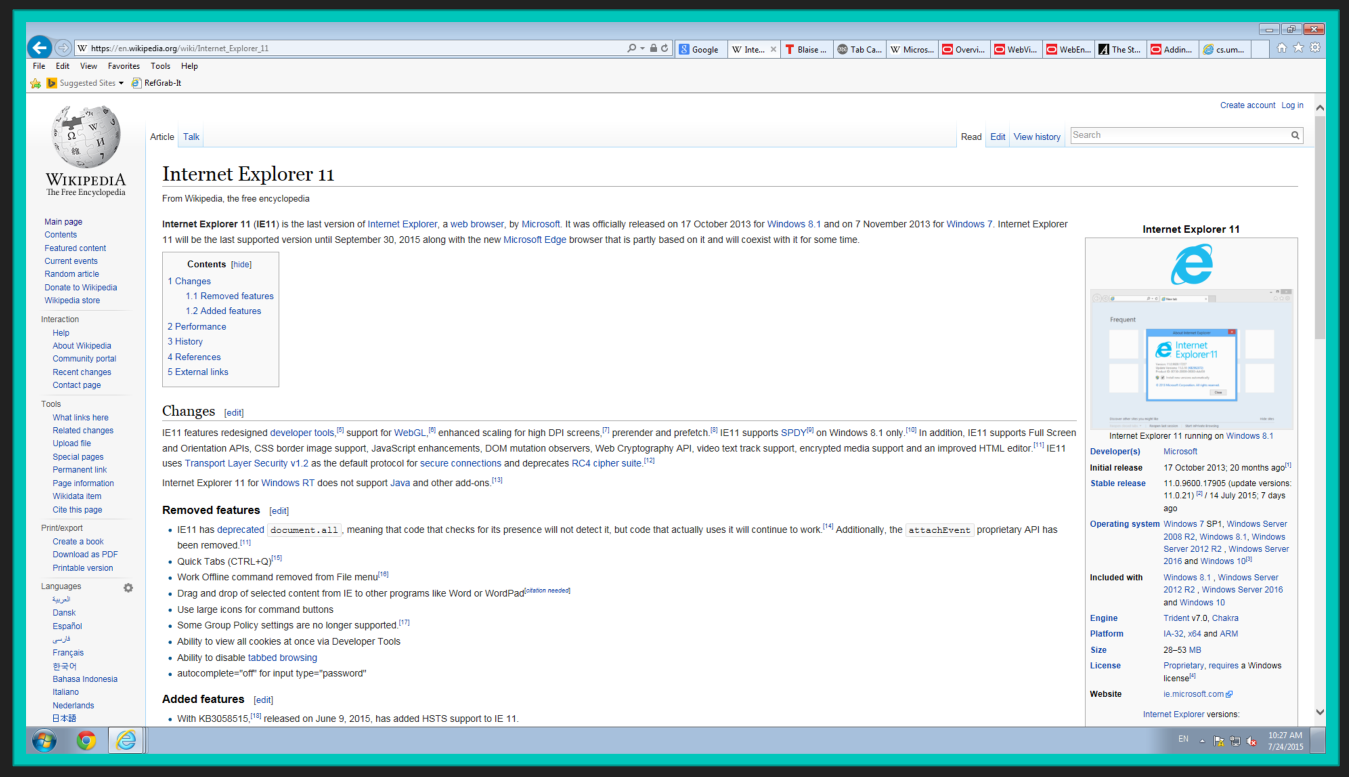
Development Process

- Refining the !knot user interface design
- Learning JavaFX and its event structure
- Implementing touch gestures
- Handling concurrency of touch events with semaphores
- Troubleshooting and debugging
- Adapting controls to unforeseen challenges
- Coding with future changes in mind

!knot v.014



Internet Explorer 11



Acknowledgements

Thank you to the St. Lawrence University Fellowship Program, Daniel F. '65 and Ann H. Sullivan Endowment for Student/Faculty Research, and Jack and Sylvia Burry University Fellows Endowment for funding this research project.

Current Features

- QUICK SLIDE: Pan all nodes
- LONG HOLD NODE: Set the node as the focus
- DELETE: close the focused tab
- LONG HOLD DRAG ON NODE: Relocate that node
- PINCH ZOOM: Zoom all if no focus; zoom focus if focus
- TAB: Toggle Global and Local Controls (for focused)
- LONG HOLD OVER NO NODE / ESCAPE: Remove focus of nodes
- WITH URL IN TOP BAR
  - WITH FOCUS
    - ENTER: Change that node to that page
  - IF NOT IN LOCAL CONTROLS
    - HOLD SUBMIT AND DRAG: Spawn a new node at the released location

Future Works

Implementing manual and automatic linking of open web documents in addition to various other features.

Further Reading

[1] Anthony, S. (2014, January 8). *Photosynth 3D: Microsoft's plan to turn the entire world into and explorable 3D panorama.* Retrieved from Extreme Tech.
[2] Bederson, B. B. (2011). The promise of zoomable user interfaces. *Behavior & Information Technology, 30*, 853-866.
[3] Bernstein, J. (2014, May 9). *Meet The Man Who Invented The Browser Tab*. Retrieved from BuzzFeed News.
[4] Chokshi, N. (2010, September 2). *The Stagnant History of the Browser Interface: A Retrospective*. Retrieved from The Atlantic.
[5] *Internet Explorer 7* (2015, July 28). Retrieved from Wikipedia.
[6] Markham, K. M., & Mintzes, J. J. (1994). The concept map as a research and evaluation tool: Further evidence of validity. *Journal of Research in Science Teaching, 31*, 91-101.
[7] Mosaic Versions (2013, December 3). Retrieved from The National Center for Supercomputing Applications.
[8] *NCSA Mosaic* (2013, December 3). Retrieved from The National Center for Supercomputing Applications.
[9] *Omniweb* (2015, April 28). Retrieved from Wikipedia.
[10] Seymour, Jim (1994). "Everything Old Is New Again". *PC Mag.* pp. 99–100.
[11] Volovyk, V. (2012, July 21). *Tabs Outliner*. Retrieved from Chrome Web Store.
[12] Wigdor, D., & Wixon, D. (2011). *Brave NUI World: Designing Natural User Interfaces for Touch and Gesture.* Burlington, MA: Morgan Kaufmann.